

QUESTIONS 9/9 completed

Factorial-Fun. ▾

FACTORIAL-FUN.

Write a function to compute the factorial of a given integer.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 long long factorial(int n) {
4     if (n < 0) return -1;
5     long long result = 1;
6     for (int i = 2; i <= n; i++) {
7         result *= i;
8     }
9     return result;
10 }
11 int main() {
12     int n;
13     cin >> n;
14     long long ans = factorial(n);
15     if (ans == -1) {
16         cout << "Invalid input";
17     } else {
18         cout << ans;
19     }
20     return 0;
21 }
```

OUTPUT

120

QUESTIONS 9/9 completed

Prime-Fun. ▾

PRIME-FUN.

Write a function to determine if a given integer is a prime number or not.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 bool isPrime(int n) {
4     if (n < 2) return false;
5     for (int i = 2; i < n; i++) {
6         if (n % i == 0) return false;
7     }
8     return true;
9 }
10 int main() {
11     int n;
12     cout << "Enter a number: ";
13     cin >> n;
14
15     if (isPrime(n))
16         cout << n << " is a prime number.\n";
17     else
18         cout << n << " is not a prime number.\n";
19     return 0;
20 }
```

OUTPUT

Enter a number: 6 is not a prime number.

INPUT

6

QUESTIONS 9 / 9 completed

GCD-Fun. ▾

GCD-FUN.

Write a function to determine the GCD (greatest common divisor) of two given integers.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int gcd(int a, int b) {
4     if (b == 0)
5         return a;
6     return gcd(b, a % b);
7 }
8 int main() {
9     int a, b;
10    cout << "Enter two integers: ";
11    cin >> a >> b;
12    cout << "GCD of " << a << " and " << b << " is: " << gcd(a, b) << endl;
13    return 0;
14 }
```

OUTPUT

```
Enter two integers: GCD of 12 and 18 is: 6
```

INPUT

12 18

QUESTIONS 9 / 9 completed

Palindrome-Fun. ▾

PALINDROME-FUN.

Write a function to check if a given string is a palindrome or not.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 bool isPalindrome(string s) {
4     int i = 0, j = (int)s.size() - 1;
5     while (i < j) {
6         if (s[i] != s[j]) return false;
7         i++;
8         j--;
9     }
10    return true;
11 }
12 int main() {
13    string str;
14    getline(cin, str);
15    if (isPalindrome(str))
16        cout << "Palindrome";
17    else
18        cout << "Not Palindrome";
19    return 0;
20 }
```

OUTPUT

```
Not Palindrome
```

INPUT

hello

QUESTIONS 9 / 9 completed

Area ▾

AREA

Write a function to calculate the area of a circle given its radius.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     double r,area;
5     cin>>r;
6     area=3.14*r;
7     cout<<area;
8 }
```

OUTPUT

```
12.56
```

INPUT

```
4
```